

D.EL.ED FIRST SEMESTER • PAPER S1.6

Pedagogy of Mathematics - I

Foundational Numeracy, Concrete Models & Mathematical Discourse • 2026 Session

EXAM BREAKDOWN

35 Ext + 15 Int

Total: 50 Marks

COURSELOAD

2 Credits

50 Study Hours

PEDAGOGY OF MATHEMATICS - I

Comprehensive Study Notes on Foundational Numeracy (FLN), Piaget's Epistemology, ELPS & Pedagogical Models, Pre-Number Concepts, Core Competency Architecture, 360° PARAKH Assessment, and Master Mathematicians

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● Curriculum Session 2026

**SYLLABUS BLUEPRINT & MARKS ALLOCATION**

5 Core Units • 35 External + 15 Internal Marks

Unit 1

8 Marks

Intro to Early Math & FLN

NIPUN Mission, Foundational Numeracy, Learning Outcomes

Unit 2

7 Marks

Principles & Approaches

Piaget, Error Diagnostics, ELPS, REACT, 4-Block

Unit 3

10 Marks

Core Math Components

Pre-Number, Place Value, Spatial Shapes, Data

Unit 4

5 Marks

Assessment in Math

PARAKH, 360° Evaluation, Holistic Report Card

Unit 5

5 Marks

Great Mathematicians

Aryabhata, Brahmagupta, Ramanujan, Euclid, Pythagoras

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UNIT 1: INTRODUCTION TO PREPARATIONAL & FOUNDATIONAL MATHEMATICS

FLN Imperatives, Meaning of Foundational Numeracy, Early Math Domains & Competency Learning Outcomes

WEIGHTAGE: 8 MARKS

1.1 & 1.2 FOUNDATIONAL LITERACY & NUMERACY (FLN): THE NATIONAL MISSION

NEP 2020 & NIPUN Bharat Mandate

The National Education Policy (NEP 2020) identifies universal acquisition of **Foundational Literacy and Numeracy (FLN)** by Grade 3 as an urgent, non-negotiable national priority. Empirical research in cognitive neuro-development shows that when children fail to acquire basic reading comprehension and number sense by the age of 8 (Grade 3), they suffer from a severe *cumulative learning deficit*. They can no longer "learn to read and calculate," rendering future curricular learning in science, social studies, and advanced mathematics largely inaccessible.

**The Learning Crisis**

National surveys (NAS, ASER) revealed that over 5 crore elementary students could not perform simple 2-digit subtraction with borrowing or identify two-digit numbers, creating a severe foundation gap.

**NIPUN Bharat Mission**

Launched under Samagra Shiksha: *National Initiative for Proficiency in Reading with Understanding and Numeracy* sets clear grade-wise milestones (Lakshyas) to be achieved across India.

**Economic & Cognitive Dividend**

Mastering foundational numeracy unlocks higher-order reasoning, logical problem-solving, algorithmic thinking, and vocational productivity required in a 21st-century knowledge economy.

1.3 & 1.4 DECONSTRUCTING FOUNDATIONAL NUMERACY (FN)

Number Sense vs. Mechanical Calculation

Foundational Numeracy (FN) is not the ability to mechanically chant multiplication tables or copy algorithmic additions from a blackboard. It is the ability to reason, solve quantitative problems in daily life, and comfortably navigate numbers, shapes, and measurements. It represents an integrated psychological ecosystem anchored in **Number Sense**:

KEY DIMENSION	TRADITIONAL ROTE COMPUTATION (যান্ত্রিক গণনা)	TRUE FOUNDATIONAL NUMERACY (সংখ্যা বোধ)
Cognitive Goal	Memorizing procedures, facts, and steps without understanding the underlying logic.	Intuitive understanding of numbers, their relationships, and multiple decomposition pathways.
Handling Addition (e.g., $18 + 7$)	Stacking numbers vertically in columns; struggling if carry-over rules are forgotten.	Flexible mental bridging: $18 + 2 = 20$, then $20 + 5 = 25$; or $10 + (8 + 7) = 25$.
Contextual Application	Fails to translate textbook sums into real-life buying, selling, or estimation.	Effortlessly estimates change in Indian Rupees (₹) at the local shop, compares vegetable prices, and measures distances.
Affective State	Causes acute <i>mathematics anxiety</i> (Math Phobia) and feeling of helplessness.	Builds mathematical confidence, joyful curiosity, and persistent problem-solving drive.

1.5 & 1.6 MAJOR ASPECTS OF EARLY MATHEMATICS & COMPETENCY LEARNING OUTCOMES

NCF-FS 2022 & NIPUN Lakshya

1. Pre-Number Concepts Matching, sorting, classifying, comparing size/length, ordering, and establishing one-to-one correspondence before symbols.	2. Numbers & Operations Counting, subitizing, place value understanding (10 ones = 1 ten), addition, subtraction, multiplication as repeated addition.	3. Shapes & Spatial Thinking Identifying 2D and 3D shapes in the environment, spatial language (above, below, inside, outside), tangrams, and symmetry.
4. Measurement & Money Transitioning from intuitive non-standard units (handspans, paces) to standard units for length, weight, volume, time, and money in Rupees (₹).	5. Patterns & Early Algebra Observing, extending, and creating repeating and growing patterns with beads, leaves, sounds, and numbers (such as 2, 4, 6, 8, ...).	6. Data Handling & Communication Collecting information, drawing simple pictographs, tallying objects, and orally articulating mathematical reasoning with peers.

Understanding Learning Outcomes (LOs): An LO is an explicit, observable, and measurable statement of what a learner is expected to know, understand, and demonstrate after a learning experience. Unlike traditional syllabus completion ("covered Chapter 3"), LOs focus on masterable student competencies (e.g., "Class 2 child can read and write numerals up to 99 and solve real-life addition problems up to 99 without carry-over").

2.1 & 2.2 UNDERSTANDING THE INDIVIDUAL LEARNER & JEAN PIAGET'S EPISTEMOLOGY

Constructivist Engine

Young children enter primary school with diverse informal mathematical knowledge rooted in their cultural home experiences. However, their cognitive apparatus processes mathematical reality very differently from adult minds. As Jean Piaget demonstrated, abstract mathematical logic cannot be transmitted verbally into a child's head; it must be constructed through physical actions on real objects:

1. Preoperational Stage (Ages 2–7)

Child relies on perceptual appearance (centration). If 5 counters are spread out, the child insists there are "more counters" because the row is longer.

Irreversibility prevents mental back-tracking ($4 + 3 = 7$ implies $7 - 3 = 4$).

2. Conservation Milestone (~Ages 6–7)

The vital watershed in math learning! Conservation of number, length, and mass means understanding that quantity remains unaltered despite physical rearrangements. Formal arithmetic fails before conservation matures.

3. Concrete Operational (Ages 7–11)

Logical thought blossoms, but remains tied strictly to **concrete, tangible objects**. Abstract symbols (x, y , formulas) cause acute confusion unless preceded by physical manipulative experiences.

2.3 MATHEMATICAL ERRORS AS WINDOWS INTO CHILDREN'S THINKING

Error Diagnostic Pedagogy

Traditional authoritarian teaching treats student errors as punishable mistakes or evidence of carelessness. Modern constructivist pedagogy views errors as **valuable diagnostic windows** revealing the child's internal alternative conceptual logic. Children rarely make random guesses; they often invent consistent, "buggy rules" when they lack underlying place-value understanding.

CHILD'S SYSTEMATIC MISCONCEPTION

Buggy Rule

Tens
Ones
5
2
- 2
7
3
5

What went wrong in the child's mind?

In the **Ones column**, the problem requires subtracting 7 from 2 (which cannot be done with whole positive counters without borrowing).

Instead of regrouping from the Tens, the child invents a shortcut rule: "Whenever subtracting, always take the smaller digit from the larger digit!" So they calculate: $7 - 2 = 5$.

Next, in the **Tens column**, since they did not borrow, they compute $5 - 2 = 3$.
Erroneous Answer: 35 (Incorrect by 10!).

Diagnosed Bug: "Smaller-from-Larger" Reversal (No Regrouping Schema)

CORRECT PLACE-VALUE REGROUPING

Sound Understanding

Tens
Ones
(4)
(12)
5
2
- 2
7
2
5

Step-by-Step Regrouping:

- You have 2 ones and need to subtract 7. Since $2 < 7$, you must borrow 1 ten from the 5 tens.
- 5 tens becomes 4 tens.
- The borrowed 1 ten is 10 ones. Add to the 2 ones: $10 + 2 = 12$ ones.
- Now subtract in the Ones column: $12 - 7 = 5$.
- In the Tens column: $4 - 2 = 2$.

Teacher's Constructivist Remediation:

- Never force mechanical drilling of the algorithm 20 times in their notebook!
- Use Bundles & Sticks:** Give the child 5 bundles of 10 matchsticks and 2 loose sticks (52).
- Prompt with a question:** "Can you give me 7 loose sticks? We only have 2 loose sticks on the desk. What can we do to one of our bundles of 10?"
- The child physically unties 1 bundle into 10 loose sticks, combines them with the 2 ($10 + 2 = 12$), removes 7, leaving 5 loose sticks and 2 bundles. Answer: 25. The misconception disappears forever!

Correct Answer: $52 - 27 = 25$ • Concrete Experience Resolves the Bug

2.4 INNOVATIVE PEDAGOGICAL FRAMEWORKS FOR EARLY MATHEMATICS

ELPS, REACT & The 4-Block Model

THE GOLD STANDARD: PAMELA LIEBECK'S ELPS APPROACH ($E \rightarrow L \rightarrow P \rightarrow S$)

Universal Early Math Sequence

Mathematical abstractions cannot be taught directly through abstract symbols. Children must travel through four sequential developmental stages:

E

Experience

Hands-on manipulation with concrete objects (pebbles, leaves, beads, sticks).

L

Spoken Language

Describing the action in mother tongue: "I had 3 mangoes and added 2 more".

P

Pictures

Representing quantity through drawings, pictographs, or dot diagrams.

S

Symbols

Writing abstract mathematical numerals and operational signs: $3 + 2 = 5$.

1. REACT Approach (CORD)

- Relating (connecting to life experiences)
- Experiencing (hands-on exploration)
- Applying (using in daily context)
- Cooperating (peer learning)
- Transferring (novel problem-solving).

2. The 4-Block Approach

- Block 1:** Oral Math Talk & Number Sense (15 min)
- Block 2:** Core Skill Teaching & Procedures (20 min)
- Block 3:** Math Games & Activity Corners (20 min)
- Block 4:** Guided Problem Solving & Journaling (15 min).

3. Gradual Release of Responsibility (GRR)

- "I Do" (Teacher explicit modeling)
- "We Do" (Guided collaborative practice)
- "You Do Together" (Peer group activity)
- "You Do Alone" (Independent confident mastery).

Art, Toy & Sports Integration: Using native bamboo toys, weaving patterns, seed counting games, Hopscotch (Ek-dukhi) for number line jumps, and Rangoli geometric symmetry to eliminate math phobia!

3.1 PRE-NUMBER CONCEPTS: THE COGNITIVE PRECONDITIONS FOR COUNTING

Foundational Pre-Requisite

A common mistake in elementary schooling is forcing children to memorize digits (1, 2, 3, ...) on day one. Before symbols make cognitive sense, children must master six fundamental **Pre-Number Concepts** through physical object play:

1. Matching (মিল কৰা)

Pairing identical or complementary items (e.g., matching a cup to a saucer, a lock to a key, or a leaf to its outline).

2. Sorting & Classification (শ্রেণীবিভাজন)

Grouping a collection of varied objects by a single common attribute: color (all red buttons), size (big leaves), or shape.

3. Comparing (তুলনা)

Developing perceptual vocabulary: big/small, heavy/light, long/short, more/less, tall/short, thick/thin.

4. Ordering / Seriation (ক্রমিক সজ্জা)

Arranging a series of objects in ascending or descending sequence based on graded magnitude (e.g., arranging 5 sticks from shortest to longest).

5. One-to-One Correspondence (একৈক সম্পর্ক)

Pairing each member of one set with exactly one member of another set (e.g., matching 1 bottle to 1 cap). The biological cornerstone of true counting!

6. Pattern Recognition (বিন্যাস চিনাক্তকরণ)

Identifying repeating rhythmic visual orders (such as circle, triangle, circle, triangle...), forming the cognitive roots of early algebraic thought.

3.2 NUMBERS, COUNTING NUANCES & THE FOUR BASIC OPERATIONS

Number Architecture

Counting Principles & Subitizing

- **Subitizing:** Instantly perceiving the quantity of a small group of items without counting (e.g., seeing 5 dots on a die and shouting "five!").
- **Stable Order Principle:** Number words must always be recited in the same fixed order (1, 2, 3, 4, ...).
- **Cardinality Principle:** The final number name counted represents the total quantity of the entire set!
- **Abstraction & Order Irrelevance:** Any collection can be counted regardless of object type or spatial arrangement.

Place Value Architecture & Operations

- **Place Value:** Built using bundle-and-stick methods or Base-10 blocks (10 units = 1 ten; 10 tens = 1 hundred).
- **Addition:** Combining sets or augmenting a quantity ($5 + 3 = 8$).
- **Subtraction:** Taking away, finding the missing part, or comparison/difference ($8 - 3 = 5$).
- **Multiplication:** Conceptualized as **repeated addition** ($3 \times 4 = 4 + 4 + 4 = 12$) or arrays.
- **Division:** Conceptualized as **equal sharing** ($12 \div 3 = 4$) or repeated subtraction.

3.3 – 3.7 SPATIAL GEOMETRY, MEASUREMENT PROGRESSION, PATTERNS & DATA

Integrated Mathematical Literacy

Shapes & Space

Exploring 3D solids (boxes, balls, cans) before abstract 2D flat shapes (squares, circles). Playing with the ancient 7-piece **Tangram** puzzle and creating floor tessellations.

Measurement Arc

Stage 1: Direct comparison (holding sticks side-by-side). Stage 2: Non-standard units (handspans, footsteps, cups of water). Stage 3: Realizing need for uniform standard metric units (meters, centimeters, kilograms, liters) and currency (₹).

Patterns & Algebra

Progressing from repeating sensory patterns (clap, tap, clap, tap) to geometric growth patterns and numerical rules (5, 10, 15, 20, ...). Patterns cultivate predictive mathematical reasoning.

Data & Communication

Children collect favorite fruit data of peers, use concrete tally marks, create colorful visual **pictographs**, and orally articulate inferences: "Most classmates love mangoes!"

4.1 – 4.3 CRITERIA FOR ASSESSMENT, PARAKH & THE HOLISTIC PROGRESS CARD (HPC)

Zero Formal Tests

In early mathematics, formal sit-down pen-and-paper examinations, percentage rankings, and pass-fail marks are strictly prohibited under NEP 2020 and NCF-FS 2022. Assessment is conducted **for learning** and **as learning**, seamlessly woven into joyful daily classroom interactions:

PARAKH 360° MULTIDIMENSIONAL ASSESSMENT MODEL

PARAKH (*Performance Assessment, Review, and Analysis of Knowledge for Holistic Development*) shifts evaluation away from rote recall toward competency mastery by integrating multiple feedback voices:

- **Self-Assessment:** The child reflects on math activities using smiley stamps (e.g., "I loved building shapes with sticks today!").
- **Peer Assessment:** Collaborative group observation during math games, puzzles, and pattern building.
- **Teacher Observation:** Continuous narrative logs tracking student milestones, spatial vocabulary, and mathematical perseverance.
- **Parental Feedback:** Insights into child application of numbers and currency change in home tasks.

THE HOLISTIC PROGRESS CARD (HPC) FOR FOUNDATIONAL NUMERACY

Instead of recording numerical marks (e.g., "35 out of 50 marks"), the HPC provides descriptive developmental feedback across 3 proficiency tiers:

Level 1 (Beginning): Needs concrete physical support to count and compare quantities.

Level 2 (Progressing): Confidently performs operations with partial pictorial scaffolding.

Level 3 (Proficient): Independently applies number sense to solve daily math problems.

Never rank or label a foundational child; assess to guide growth!



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Prepared by Sayeem Sadik

1. ARYABHATA (476–550 CE)

Ancient India

The luminous astronomer-mathematician from Kusumapura (Pataliputra), author of the celebrated treatise **Aryabhatiya**:

- **Zero & Place Value**: Formalized the decimal place-value notation where zero served as both symbol and place-holder.
- **Approximation of π** : Calculated $\pi \approx \frac{62832}{20000} \approx 3.1416$, noting explicitly that it was an irrational approximation!
- **Trigonometry & Astronomy**: Formulated foundational sine tables (*Jya*) and proved Earth rotates daily on its axis and revolves around the Sun.

Author of *Aryabhatiya* • Pioneer of Decimal Notation

2. BRAHMAGUPTA (598–668 CE)

Bhinmal, Rajasthan

The mathematical titan who headed the observatory at Ujjain, author of **Brahmasphutasiddhanta**:

- **Rules of Zero (*Shunya*)**: First in human history to treat zero as a full operational number with algebraic rules: $0 + a = a$, $a - 0 = a$, $a \times 0 = 0$.
- **Negative Numbers**: Defined operational arithmetic of negative numbers using debts (*rina*) and fortunes (*dhana*).
- **Cyclic Quadrilaterals**: Discovered the monumental formula for the area of a cyclic quadrilateral: $\text{Area} = \sqrt{(s-a)(s-b)(s-c)(s-d)}$, where $s = \frac{a+b+c+d}{2}$.

The Father of Zero Arithmetic & Negative Numbers

3. SRINMASA RAMANUJAN (1887–1920)

Erode, Tamil Nadu

The self-taught mathematical genius whose intuitive brilliance astounded Cambridge Professor G.H. Hardy:

- **Number Theory & Partitions**: Breakthrough formulas for integer partitions $p(n)$ and mock theta functions.
- **The Hardy–Ramanujan Number (1729)**: Smallest integer expressible as the sum of two cubes in two different ways: $1729 = 1^3 + 12^3 = 9^3 + 10^3$.
- **National Mathematics Day**: December 22 celebrated annually across India in honor of his birth anniversary.

Taxicab Number 1729 • Celebrated Dec 22

4. EUCLID OF ALEXANDRIA (C. 300 BCE)

Ancient Greece

Universally celebrated as the "**Father of Geometry**" who revolutionized human logical deduction:

- **The Elements (*Stoicheia*)**: The world's most influential mathematics textbook spanning 13 volumes.
- **Axiomatic Method**: Proved complex geometric theorems rigorously from a foundational base of 5 common notions and 5 postulates (including the Parallel Postulate).
- **Number Theory**: Proved there are infinitely many prime numbers and developed the Euclidean Algorithm for the Greatest Common Divisor (GCD/HCF).

Author of *The Elements* • Father of Axiomatic Geometry

5. PYTHAGORAS OF SAMOS (C. 570–495 BCE)

Ancient Greece

Philosopher-mathematician who established the Pythagorean brotherhood, viewing numbers as the cosmic essence of reality:

The Pythagorean Theorem:
In any right-angled triangle, the square of the hypotenuse equals the sum of the squares of the other two sides:

$$a^2 + b^2 = c^2$$

Identified integer triples: (3, 4, 5), (5, 12, 13), (8, 15, 17).

Discovery of Incommensurability (Irrational Numbers):
Proved that the diagonal of a unit square ($\sqrt{2}$) cannot be expressed as a ratio of two integers ($\frac{p}{q}$, such as $\frac{5}{3}$), discovering irrational numbers! Also connected mathematical string ratios to musical acoustics (octaves).

$a^2 + b^2 = c^2$ • Discovery of Irrational Numbers ($\sqrt{2}$)

★ PAPER S1.6 PRACTICUM & INTERNAL ASSESSMENT ARCHITECTURE (15 MARKS)

Syllabus 2026

Compulsory Activity

3 Marks

Class 1–3 Lesson Analysis

Prescribed Project

3 Marks

Survey / Diagnostic Study

Assignments (Any 2)

3 Marks

TLM / Rangoli / Symmetry

Institutional Unit Test

6 Marks

Internal Written Evaluation

Prescribed Practicum Activity Options:

- **Compulsory Task (3 Marks)**: Select any mathematics lesson from Class 1 to 3 SCERT textbooks; analyze it in terms of target Learning Outcomes (LO), design concrete learning activities using the ELPS approach, and develop an innovative instructional aid (TLM).
- **Project Option 1**: Select a village neighborhood; survey educational status across households whose house numbers are multiples of 3 (3, 6, 9, 12, . . .); compile statistical bar diagrams and interpretive inferences.
- **Project Option 2**: Visit five local weekly markets; survey and compare the prices of 15 common vegetables in Indian Rupees (₹); present statistically valid charts and cost-saving inferences.
- **Project Option 3**: Draw an illustrated developmental progression chart showing the stages of child growth and mapping expected mathematical milestones according to Jean Piaget's theory.
- **Assignments Option**: Create mathematical Rangoli designs illustrating geometric symmetries; or prepare a symmetrical classification chart dividing English alphabet letters (A, B, C, . . .) and numerals (0, . . . , 9) along lines of symmetry.

School Internship (SIP-I Linkage):
During SIP-I, student-teachers will map foundational numeracy competencies in primary classrooms, set up an interactive "Math Learning Corner" containing concrete manipulatives (pebbles, beads, tangrams, dice), deliver ELPS-aligned learning episodes, and prepare a reflective analytical report countersigned by the school head.

★ Paper S1.6 Study Notes • Curriculum Session 2026

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